Mode S & ADS-B Statistics

What the receiver hears from the aircraft, and how to read and use the Statistics page

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In short. Every wind, temperature and windshear value of MODE-S Wind comes from radio messages that the aircraft transmit on 1 090 MHz: replies to the interrogations of the ground radars and the wide-area multilateration (WAM) network, and the aircraft's own ADS-B broadcasts. The Statistics page counts these messages as the receiver decodes them and shows, refreshed every 10 seconds:

- **what arrives** — messages per second, the downlink formats, the Comm-B registers and the ADS-B message types;
- **how often each aircraft answers** — the reply cadence of the registers that give the wind, per flight phase: the sample rate behind every wind and windshear value;
- **who is asking** — the ground interrogators identified from the all-call replies;
- **what it gives** — the winds computed per minute, and the positions in the Radarcape aircraft list.

Section 1 explains Mode S and ADS-B as far as needed to read the page, Section 2 the page itself, and Section 3 what can be analysed with it.

A passive view. The receiver hears the aircraft (1 090 MHz), not the ground stations (1 030 MHz). Everything on the page is inferred from the replies that reached the antenna in Tuusula; the counts start again when the server restarts. The figures in this guide show the page with *simulated* traffic — the layout is the real one, the numbers on your screen will differ.

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1 Mode S and ADS-B in a nutshell

1.1 Secondary surveillance radar and Mode S

Interrogation and reply. Secondary surveillance radar (SSR) does not see the aircraft by its echo: a ground station *interrogates* on 1 030 MHz and the aircraft's *transponder* replies on 1 090 MHz. The old modes A and C answer every interrogation with the identity code (squawk) or the pressure altitude; with many aircraft their replies overlap (*garbling*) and a radar also receives replies meant for other radars (*FRUIT*, false replies unsynchronised in time).

Mode S ("select") solves this by giving every aircraft a unique **24-bit ICAO address**. A ground station first finds new aircraft with an *all-call* interrogation, which every Mode S transponder answers with its address; then it asks each aircraft individually (*roll-call*), addressed to it alone [1, 3]. A station that has acquired an aircraft can *lock it out* of its own further all-calls, so that the aircraft does not keep answering them. Each station identifies itself in the all-call by an **interrogator code**: an *II code* (interrogator identifier, 0–15) or an *SI code* (surveillance identifier, 1–63, added in 1997 for denser networks) [5].

Squitters. Without being asked, a Mode S transponder also broadcasts its address about once a second (the *acquisition squitter*), and an ADS-B transponder broadcasts its position, velocity, identification and status several times a second (the *extended squitter*, Section 1.4). Anyone can receive these, including other aircraft (ACAS / TCAS) and passive receivers like the Radarcape.

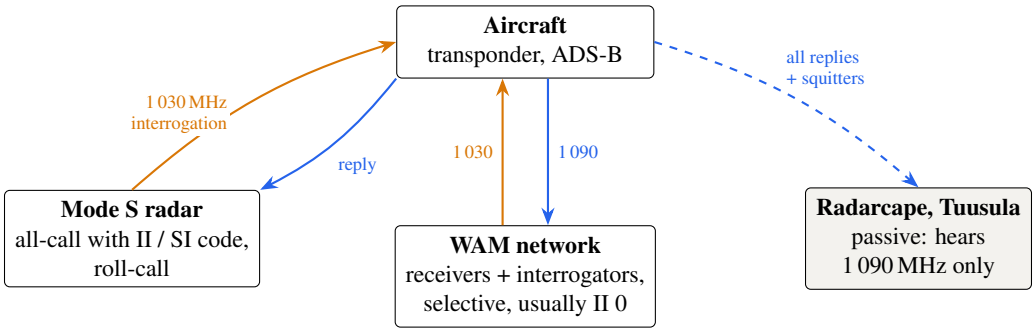


Figure 1. Who talks to whom. The ground stations interrogate on 1 030 MHz; the aircraft answers on 1 090 MHz and also broadcasts squitters. The Radarcape hears every reply and squitter that reaches Tuusula, whoever asked — but never the interrogations.

Around Helsinki the interrogations come mostly from Fintraffic's nationwide **wide-area multilateration (WAM)** network, together with Mode S radars [9, 10]. WAM locates the aircraft from the arrival times of its signals at several receivers; it also interrogates the aircraft selectively, as often as configured, and is not tied to the rotation of a radar antenna. Active WAM systems generally use II 0 [5]. The rate at which WAM asks an aircraft for its registers is the sample rate of the wind data of MODE-S Wind (Section 2.6).

1.2 The messages: downlink formats

A Mode S message is 56 bits (*short*, 64 µs with its preamble) or 112 bits (*long*, 120 µs), sent at 1 Mbit/s. The first five bits are the **downlink format** (DF); the last 24 bits are **parity** (Fig. 2).

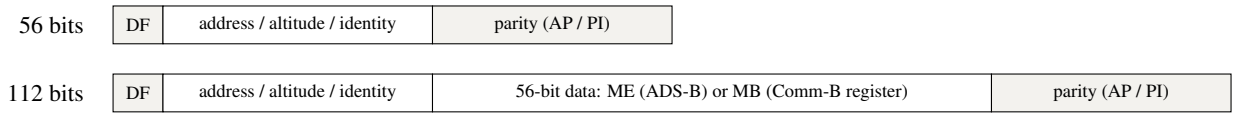


Figure 2. The two message lengths. In a long message the 56-bit data field carries one ADS-B message (DF17/18, field ME) or the content of one Comm-B register (DF20/21, field MB).

DF	Length	Content	Parity
0	short	Short air–air surveillance: reply to another aircraft's ACAS (TCAS)	AP
4	short	Surveillance reply, altitude (to a ground roll-call)	AP
5	short	Surveillance reply, identity (squawk)	AP
11	short	All-call reply, or the spontaneous acquisition squitter	PI
16	long	Long air–air surveillance (ACAS coordination)	AP
17	long	Extended squitter: ADS-B from a transponder	PI
18	long	Extended squitter from a non-transponder device; TIS-B, ADS-R	PI
19	long	Military extended squitter	—
20	long	Comm-B reply, altitude + 56-bit register content	AP
21	long	Comm-B reply, identity + 56-bit register content	AP
24	long	Comm-D extended length message	AP

Table 1. The downlink formats counted on the page.

Parity, and why some addresses cannot be verified. The parity is a 24-bit cyclic redundancy check (CRC) of the message. In **PI** formats (DF11/17/18) the CRC of a correct message is zero — or, in an all-call reply, the code of the interrogator that asked (Section 2.7). In **AP** formats (address/parity: DF0/4/5/16/20/21) the CRC is combined with the aircraft's address: the receiver recovers the address *from* the CRC, so a corrupted message simply yields a wrong address. The decoder (pyModeS [4]) therefore marks an AP reply as *verified* only when its address belongs to an aircraft already heard in its own squitters; the share of unverified Comm-B replies is shown on the page.

1.3 Comm-B and the registers

Ground-initiated Comm-B (GICB). A roll-call interrogation can ask the transponder for one of its 256 *registers* (Comm-B data selectors, BDS, written as two hexadecimal digits, e.g. 5,0). The transponder fills the registers from the flight computers and answers with a DF20 or DF21 reply carrying the 56 bits of that register (field MB). The reply does *not* say which register it is: the receiving ground station knows what it asked, a passive receiver does not. pyModeS

therefore infers the register from the content — each register has fixed status bits, value ranges and consistency rules — and when the bits fit more than one register it picks the most likely one; the page shows the share of such *ambiguous* inferences.

ELS and EHS. Europe requires *Elementary Surveillance* (ELS: registers 1,0, 1,7, 2,0 and 3,0) for IFR flights, and *Enhanced Surveillance* (EHS: 4,0, 5,0, 6,0) together with ADS-B Out for aircraft heavier than 5 700 kg or faster than 250 kt [6]. Practically all airliners therefore answer the EHS registers, which carry the true airspeed, Mach number and heading that MODE-S Wind turns into wind and temperature [11].

BDS	Content (resolution)	Used by MODE-S Wind
1,0	Data link capability: transponder level, ACAS, Mode S subnetwork version	—
1,7	Common-usage GICB capability: which registers the aircraft supports	—
2,0	Aircraft identification: the callsign, 8 characters	callsign
3,0	ACAS active resolution advisory	—
4,0	Selected vertical intention: MCP / FCU selected altitude and FMS selected altitude (16 ft), barometric pressure setting (0.1 hPa)	—
4,4	Meteorological routine air report (MRAR): wind, static air temperature, pressure, turbulence, humidity — optional, rare	direct meteo
4,5	Meteorological hazard report (MHR): turbulence, wind shear, microburst, icing, wake vortex — optional, rare	hazards
5,0	Track and turn report: roll angle (0.18°), true track (0.18°), ground speed (2 kt), track angle rate, true airspeed (2 kt)	wind, windshear
6,0	Heading and speed report: magnetic heading (0.18°), indicated airspeed (1 kt), Mach (0.004), barometric and inertial vertical rates (32 ft/min)	wind, temperature

Table 2. The Comm-B registers counted on the page [2, 3]. The wind is the ground-velocity vector of 5,0 minus the air-velocity vector from the true airspeed and the heading of 6,0; the temperature follows from the Mach number and the true airspeed.

1.4 ADS-B: the extended squitter

An ADS-B transponder broadcasts DF17 messages by itself, without being interrogated. The first five bits of the 56-bit ME field are the **type code** (TC), which tells what the message carries (Table 3). An airborne aircraft sends its position and velocity about twice a second each, its identification about every 5 s, and its status messages every 1.25–2.5 s [3, 8].

TC	Message	Typical rate
1–4	Identification and category (callsign, wake category)	every 5 s
5–8	Surface position	on the ground
9–18	Airborne position with barometric altitude (the TC also encodes the position integrity)	2 per s
19	Airborne velocity: ground speed and track (subtypes 1–2) or airspeed and heading (3–4); vertical rate; the **GNSS − barometric altitude difference** (25 ft steps)	2 per s
20–22	Airborne position with GNSS height	rare
28	Aircraft status: emergency / priority, ACAS RA broadcast	on events
29	Target state and status: selected altitude, barometric pressure setting, autopilot modes	every 1.25 s
31	Operational status: ADS-B version, NIC / NACp supplements, capabilities	every 2.5 s

Table 3. ADS-B type codes (DF17).

Versions. The *ADS-B version* in the operational status message tells which standard the transponder follows: 0 (RTCA DO-260), 1 (DO-260A) or 2 (DO-260B / EUROCAE ED-102A, required by the European mandate) [7]. Version 2 transponders report the position accuracy and integrity in the form used by the GPS Quality pages (NACp, NIC).

The GNSS − baro difference in the velocity message is what MODE-S Wind uses for the heights of the pressure surfaces (Sounding and Wind Map guide). DF18 is the same message from a device that is not a transponder (e.g. an airport vehicle) or a rebroadcast by a ground station (TIS-B, ADS-R; mainly in the USA).

2 The Statistics page

Statistics is a link in the top bar. The page asks the server for the counts every 10 seconds (not while its browser tab is hidden). The Window buttons at the top right — 1 min, 10 min (default) and 1 h — choose the period of all tables; their rates are per second averaged over that period. The choice is remembered in the browser. Most headers, column titles

and values explain themselves when hovered.

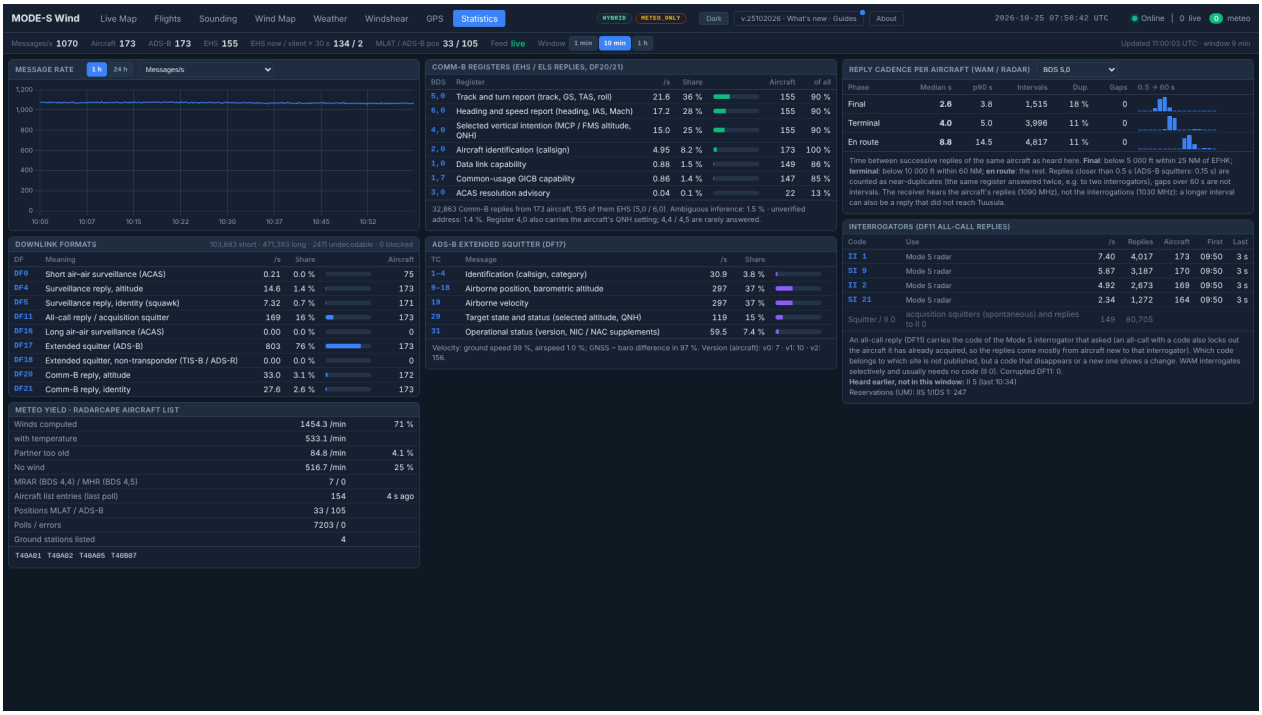


Figure 3. The Statistics page (simulated traffic). Left column: message rate, downlink formats, meteo yield and the Radarcape aircraft list. Middle: Comm-B registers, ADS-B type codes. Right: reply cadence, interrogators.

How the counting works. The collector counts every frame from the Radarcape Beast feed before decoding, then each decoded message by format, type code and register, and the time since the same aircraft's previous message of the same kind. Everything stays in memory — per-minute records for the last hour and a compact per-minute series for 24 hours — and nothing is written to the database or the SD card; after a restart the counts begin again. Counting costs a few microseconds per message.

2.1 Summary bar

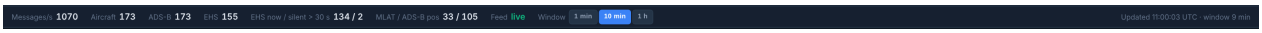


Figure 4. The summary bar.

- **Messages/s** — all frames of the last complete 10 seconds (hover: the average over the window). Typical daytime values are of the order of 1 000–2 000 per second.
- **Aircraft**, **ADS-B**, **EHS** — distinct addresses heard in the window; those that sent ADS-B (DF17); those that answered register 5,0 or 6,0.
- **EHS now / silent > 30 s** — aircraft heard in the last 10 s whose last BDS 5,0 reply is under 30 s old (being interrogated for EHS right now) / between 30 s and 10 min old (not interrogated for EHS at the moment, or their replies do not reach Tuusula). A rising "silent" count while traffic is normal points at a change in the ground system.
- **MLAT / ADS-B pos** — positions in the Radarcape aircraft list at the last poll, by source: multilateration of the Radarcape network, or the aircraft's own ADS-B.
- **Feed** — `live`, or `silent` with the time since the last frame (amber); the hover gives the connection uptime and the number of (re)connections since the server started.

2.2 Message rate

The chart shows the messages per second over the last hour (one point per 10 s) or, with `24 h`, per minute over the last day (since the server started). The selector chooses the series of the chart: messages, the main formats (DF4/5, DF11, DF17, DF20/21), the registers 5,0, 6,0, 4,0 and 2,0, the numbers of aircraft and EHS aircraft per minute, the **median BDS 5,0 interval** on final, in the terminal area and en route (Section 2.6), and the number of interrogator codes heard.



Figure 5. 24-hour views (simulated): the messages per second follow the traffic of the day; the median BDS 5,0 interval is shortest on final and longest en route, and at night, with few landings, the value on final becomes noisy.

2.3 Downlink formats

DOWNLINK FORMATS — 103,893 short · 471,393 long · 2411 undecodable · 0 blocked

DF	Meaning	/s	Share	Aircraft
DF0	Short air–air surveillance (ACAS)	0.21	0.0 %	75
DF4	Surveillance reply, altitude	14.6	1.4 %	173
DF5	Surveillance reply, identity (squawk)	7.32	0.7 %	171
DF11	All-call reply / acquisition squitter	169	16 %	173
DF16	Long air–air surveillance (ACAS)	0.00	0.0 %	0
DF17	Extended squitter (ADS-B)	803	76 %	173
DF18	Extended squitter, non-transponder (TIS-B / ADS-R)	0.00	0.0 %	0
DF20	Comm-B reply, altitude	33.0	3.1 %	172
DF21	Comm-B reply, identity	27.6	2.6 %	173

Figure 6. Downlink formats (simulated).

For each format: its meaning, the rate, the share of the decoded messages and the number of aircraft. The header gives the short and long frames, the undecodable ones (rejected by the decoder: corrupted, garbled or an unknown format) and the blocked ones (addresses on the ICAO blocklist). DF17 is usually the largest part (two position and two velocity squitters per second from every ADS-B aircraft), followed by DF11 (one acquisition squitter per second from every Mode S aircraft) and the replies to the ground interrogations (DF4/5 and DF20/21).

2.4 Comm-B registers

COMM-B REGISTERS (EHS / ELS REPLIES, DF20/21)

BDS	Register	/s	Share	Aircraft	of all
5,0	Track and turn report (track, GS, TAS, roll)	21.6	36 %	155	90 %
6,0	Heading and speed report (heading, IAS, Mach)	17.2	28 %	155	90 %
4,0	Selected vertical intention (MCP / FMS altitude, QNH)	15.0	25 %	155	90 %
2,0	Aircraft identification (callsign)	4.95	8.2 %	173	100 %
1,0	Data link capability	0.88	1.5 %	149	86 %
1,7	Common-usage GICB capability	0.86	1.4 %	147	85 %
3,0	ACAS resolution advisory	0.04	0.1 %	22	13 %

32,863 Comm-B replies from 173 aircraft, 155 of them EHS (5,0 / 6,0). Ambiguous inference: 1.5 % · unverified address: 1.4 %. Register 4,0 also carries the aircraft's QNH setting; 4,4 / 4,5 are rarely answered.

Figure 7. Comm-B registers (simulated).

Each register heard in the DF20/21 replies, with its rate, share of the Comm-B replies, the number of aircraft that answered it, and their share of all aircraft with Comm-B replies. The footer gives the share of ambiguous inferences and of replies with an unverified address (Section 1.2). The mix shows what the ground system is asking for: 5,0 and 6,0 dominate where WAM gathers EHS data; 2,0 (callsign) and 4,0 (selected altitude, used by controllers to see where an aircraft is going) come less often; 1,0 and 1,7 are asked rarely, as their content seldom changes.

2.5 ADS-B extended squitter

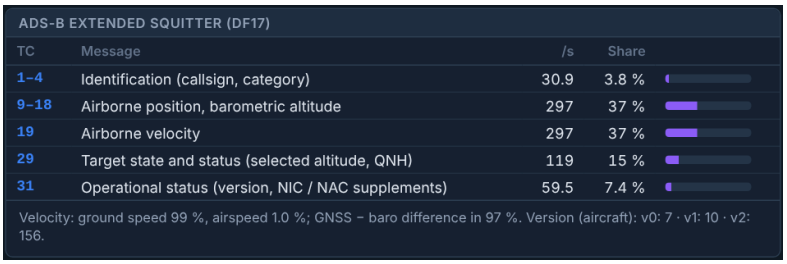


Figure 8. ADS-B type codes (simulated).

The type-code groups of Table 3 with their rates and shares. The footer gives the velocity subtypes (ground speed or airspeed), the share of velocity messages that carry the GNSS − baro difference, and the number of aircraft per ADS-B version (from TC 31).

2.6 Reply cadence per aircraft

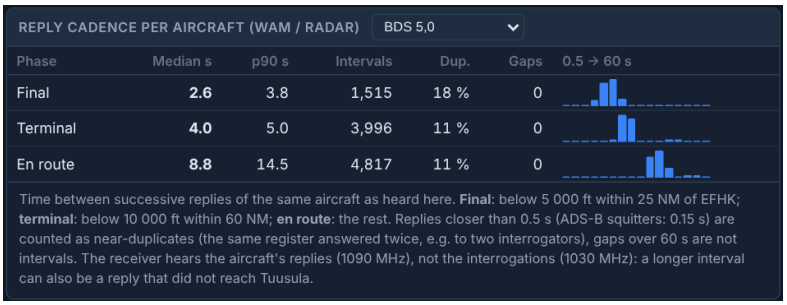


Figure 9. Reply cadence of BDS 5,0 (simulated): median, 90th percentile and histogram of the intervals per flight phase.

For the chosen register or message type — BDS 5,0, 6,0, 4,0, 2,0, the DF4/5 surveillance replies, the ADS-B position or velocity squitter — the table gives the time between successive messages of the *same aircraft*:

- per **flight phase**: **final** (below 5 000 ft within 25 NM of EFHK), **terminal** (below 10 000 ft within 60 NM), **en route** (the rest);
- **Median** and **p90** (90 % of the intervals are shorter), and a **histogram** from 0.5 to 60 s (hover a bar for its range and count);
- **Dup.** — *near-duplicates*: the same register again within 0.5 s (for the ADS-B squitters, which come every 0.4–0.6 s by themselves: within 0.15 s). They are not new samples; typically the aircraft answered two interrogators asking almost at the same time. They are left out of the intervals;
- **Gaps** — pauses of more than 60 s, not counted as intervals.

Why it matters. A wind needs a 5,0 and a 6,0 reply close in time, and the windshear detection follows TAS − GS from each 5,0 reply. The BDS 5,0 interval on final is therefore the time resolution of the approach winds and of the windshear detection: at a ground speed of 140 kt an interval of 2.5 s means one sample every 180 m. The ADS-B velocity squitter (every 0.5 s) gives the ground speed five times as often, but not the airspeed.

Reading it with care. The receiver only hears the replies. A long interval can mean that nobody asked, or that the reply did not reach Tuusula (terrain, distance, a weak or shadowed antenna on the aircraft, interference). Intervals of aircraft close to the receiver and high up are the most reliable measure of the interrogation schedule.

2.7 Interrogators

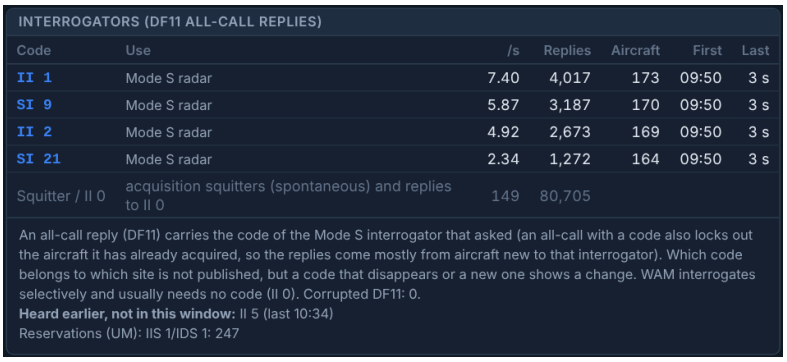


Figure 10. Interrogators (simulated codes): the codes heard in all-call replies with their use, rate and aircraft; below the table a code heard earlier but not in the last 10 minutes.

The parity of an all-call reply (DF11) is the CRC of the message combined with the code of the interrogator that asked. Computing the CRC therefore reveals **which interrogator** the aircraft answered — the only place where a passive receiver sees the ground side. A remainder of 0 is either the spontaneous acquisition squitter or a reply to II 0 (the two cannot be told apart); any other value that is not a valid code means a corrupted reply.

II	matching SI codes	Use (ICAO EUR / MID practice)
1–13	IC + 16, + 32, + 48	Operational Mode S radars; allocated by the EUROCONTROL MICA cell
0	16, 32, 48	Multilateration, WAM and mobile interrogators (not allocated)
14	14, 30, 46, 62	Test, research and development
15	15, 31, 47, 63	Military

Table 4. Interrogator codes [5]. An SI code has the same 4-bit value as its II code (e.g. II 1 and SI 1, 17, 33, 49).

For each code the table shows its use (Table 4), the reply rate, the replies in the window, the number of aircraft that replied, when the code was first heard since the server started and how long ago its last reply came. Codes heard earlier but not in the window are listed below the table. Because a radar locks out the aircraft it has acquired, its all-call replies come mostly from aircraft new to it — entering its coverage, taking off, or recently changed — so a code's rate is not a measure of how busy the radar is, but its presence is a sign that it is transmitting. Which site uses which code is not published.

The footer also sums up the **reservation subfields** of the surveillance and Comm-B replies (UM: IIS, the code of the interrogator holding a multisite reservation, and IDS, its type). They are rarely non-zero.

2.8 Meteo yield and the Radarcape aircraft list



Figure 11. Meteo yield and the Radarcape aircraft list (simulated).

For every new BDS 5,0 or 6,0 reply: **winds computed** (the other register was close enough in time and the checks passed; with a Mach/TAS temperature as well), **partner too old** (the other reply was older than `WIND_MAX_AGE_SEC`: the wind would belong to another altitude and is not stored), **no wind** (no partner within `WIND_MAX_PAIR_AGE`, or rejected by the roll, track-rate or speed checks), and the rare MRAR / MHR reports. Below them: the entries of the

Radarcape aircraft list at the last poll, its positions by source (MLAT / ADS-B), polls and errors, and the ground stations on the ICAO blocklist that the list contains (e.g. the `T40...` WAM stations, with their source and last time in the hover).

3 What you can analyse

3.1 The sample rate of the wind data

Choose BDS 5,0 in the cadence table and compare the phases. On final the median interval is the time resolution of the approach winds and of the windshear detection; en route it decides how many aircraft contribute to an hourly profile. With the `24 h` chart and the series *BDS 5,0 interval* the same values are seen through the day: a change at a fixed hour, or a step that stays, is a change of the interrogation schedule. Compare 5,0 with 6,0: if 6,0 is asked clearly less often, fewer 5,0 replies find a partner and the *no wind* share in the meteo table rises.

3.2 Changes in the ground system

- **Interrogator codes** that appear or disappear (the list below the table, or the series *interrogators heard* of the 24-h chart): a radar switched off for maintenance, a new one in service, a change of code allocation.
- **EHS silent > 30 s** rising while the traffic is normal: aircraft are not being asked for their EHS registers (or not heard), e.g. part of the WAM network out of service.
- **Near-duplicates**: their share shows how often two stations ask the same aircraft for the same register within half a second — overlapping coverage.
- **DF4/5 rate per aircraft**: the surveillance replies show how often the aircraft are interrogated at all; with the EHS cadence it separates "not interrogated" from "interrogated, but not for EHS".

3.3 The aircraft fleet

The share of EHS aircraft among the Comm-B aircraft, the ADS-B versions and the share of velocity messages with the GNSS − baro difference describe the fleet over Southern Finland: business jets and smaller aircraft answer the ELS registers but often not EHS, older transponders report ADS-B version 0 or 1. MRAR (4,4) and MHR (4,5) replies show whether any aircraft in the area deliver direct meteorological reports — at Helsinki-Vantaa practically none do, which is why MODE-S Wind computes wind and temperature from 5,0 and 6,0.

3.4 Receiver and data quality

- **Messages per second** through the day follow the traffic; a sudden drop with normal traffic points at the receiver, the antenna or the network.
- **Undecodable share** (header of the downlink formats): high values mean interference or heavy garbling.
- **Feed** state and reconnections: interruptions of the Beast connection.
- **MLAT vs ADS-B positions**: when GNSS interference removes the ADS-B positions of many aircraft, the MLAT share rises — compare with the GPS Quality and GPS Episodes pages.
- **Unverified Comm-B addresses**: mostly replies from aircraft not yet heard in their own squitters, or corrupted replies.

3.5 ACAS activity

DF0 and DF16 are replies to the collision avoidance systems of other aircraft, register 3,0 and ADS-B TC 28 carry ACAS resolution advisories. Their rates show the ACAS interrogation load in the area; a 3,0 reply with content, or a TC 28 RA broadcast, marks an actual resolution advisory.

Question	Where to look	What to expect
How often are the winds sampled on final?	Cadence, BDS 5,0, Final	median about 2–3 s
Did the interrogation schedule change?	24-h chart, *BDS 5,0 interval*	a step or an hourly pattern
Is a radar out of service?	Interrogators, *heard earlier*	a code missing for hours
Why fewer winds than usual?	Meteo yield, cadence of 6,0	*no wind* share up
Is the receiver all right?	Messages/s, undecodable, Feed	steady with traffic
GNSS interference now?	MLAT / ADS-B positions	MLAT share up

Table 5. Quick analyses.

4 Limitations

- **One receiver.** Everything is seen from Tuusula: coverage, terrain and the aircraft antenna decide what is heard. Low aircraft far away are not heard; their intervals look longer than they are.
- **Replies, not interrogations.** The schedule of the ground stations is inferred from the replies; an unanswered or lost interrogation is invisible.
- **Inferred registers.** The register of a Comm-B reply is inferred by the decoder; a small share is ambiguous or wrong.
- **Codes, not sites.** Interrogator codes are known, their sites are not published; II 0 replies cannot be separated from acquisition squitters.
- **Memory only.** The counts restart with the server; the 24-h series begins at the start.

5 Quick reference

Item	Meaning
Window	period of the tables: 1 min, 10 min, 1 h
Messages/s	frames of the last 10 s
EHS	aircraft that answered 5,0 or 6,0 in the window
EHS silent > 30 s	heard now, last 5,0 reply 30 s – 10 min ago
Ambiguous	Comm-B replies whose bits fit more than one register
Unverified	AP replies whose address is not known from squitters
Median / p90	interval between successive replies of the same aircraft
Dup.	same register again within 0.5 s (ADS-B: 0.15 s; not a new sample)
Gaps	pauses over 60 s
II / SI	interrogator codes from DF11 all-call replies
Squitter / II 0	spontaneous acquisition squitters and replies to II 0
Winds computed	new 5,0 / 6,0 replies that completed a wind
Partner too old	wind not stored: the other reply too old
API	`/api/msgstats?window=1...60`

References

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[2] ICAO. *Technical Provisions for Mode S Services and Extended Squitter* (Doc 9871), 2nd edition, with amendments.

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